

מטלה 4 שאלה 5
איוון גורבצ'וב
קישור לזיג:

https://github.com/IvanGorbache/Envy_free_room_allocation_hw4q5.git

Valuations	Rent	Allocation
[[10, 10, 10], [20, 50, 40], [52, 42, 11]]	100	Player 0 gets Room 2 with value 10, and pays 33.33 Player 1 gets Room 1 with value 50, and pays 33.33 Player 2 gets Room 0 with value 52, and pays 33.33 Expected Rent: 100 Total Rent: 100.0
[[150, 0], [140, 10]]	100	Player 0 gets Room 0 with value 150, and pays 115.00 Player 1 gets Room 1 with value 10, and pays -15.00 Expected Rent: 100 Total Rent: 100.0
[[50, 50, 50], [50, 50, 50], [50, 50, 50]]	150	Player 0 gets Room 2 with value 50, and pays 50.00 Player 1 gets Room 1 with value 50, and pays 50.00 Player 2 gets Room 0 with value 50, and pays 50.00 Expected Rent: 150 Total Rent: 150.0
[[60, 50, 50, 10], [60, 50, 30, 10], [60, 20, 50, 10], [70, 50, 50, 10]]	200	Player 0 gets Room 3 with value 10, and pays 20.00 Player 1 gets Room 1 with value 50, and pays 60.00 Player 2 gets Room 2 with value 50, and pays 60.00 Player 3 gets Room 0 with value 70, and pays 60.00 Expected Rent: 200 Total Rent: 200.0
[[60, 50, 50, 10], [60, 50, 30, 10]]	100	Player 0 gets Room 1 with value 50, and pays 45.00 Player 1 gets Room 0 with value 60, and pays 55.00 Expected Rent: 100 Total Rent: 100.0
[[0, 10, 0], [0, 0, 10], [10, 0, 0]]	10	Player 0 gets Room 1 with value 10, and pays 3.33 Player 1 gets Room 2 with value 10, and pays 3.33 Player 2 gets Room 0 with value 10, and pays 3.33 Expected Rent: 10

		Total Rent: 10.0
[[10, 0, 0], [10, 0, 0], [10, 0, 0]]	10	Player 0 gets Room 1 with value 0, and pays 0.00 Player 1 gets Room 2 with value 0, and pays 0.00 Player 2 gets Room 0 with value 10, and pays 10.00 Expected Rent: 10 Total Rent: 10.0
[[100, 0, 0], [90, 0, 0], [80, 0, 0]]	100	Player 0 gets Room 0 with value 100, and pays 93.33 Player 1 gets Room 1 with value 0, and pays 3.33 Player 2 gets Room 2 with value 0, and pays 3.33 Expected Rent: 100 Total Rent: 99.99999999999999
[[10, 20, 30], [20, 30, 40], [30, 40, 50]]	150	Player 0 gets Room 0 with value 10, and pays 40.00 Player 1 gets Room 1 with value 30, and pays 50.00 Player 2 gets Room 2 with value 50, and pays 60.00 Expected Rent: 150 Total Rent: 150.0
[[15, 25, 35], [15, 25, 35], [5, 10, 15]]	155	Player 0 gets Room 1 with value 25, and pays 51.67 Player 1 gets Room 2 with value 35, and pays 51.67 Player 2 gets Room 0 with value 5, and pays 51.67 Expected Rent: 155 Total Rent: 155.0
[[12, 7, 20], [18, 25, 10], [5, 15, 30]]	160	Player 0 gets Room 0 with value 12, and pays 53.33 Player 1 gets Room 1 with value 25, and pays 53.33 Player 2 gets Room 2 with value 30, and pays 53.33 Expected Rent: 160 Total Rent: 160.0
[[40, 30, 20, 10], [25, 35, 15, 5], [30, 10, 40, 20], [10, 50, 20, 30]]	75	Player 0 gets Room 0 with value 40, and pays 23.75 Player 1 gets Room 1 with value 35, and pays 23.75 Player 2 gets Room 2 with value 40, and pays 23.75 Player 3 gets Room 3 with value 30, and pays 3.75 Expected Rent: 75 Total Rent: 75.0

קישור לעמוד וקיפדה על האלגוריתם של Karp:

https://en.wikipedia.org/wiki/Minimum_mean_weight_cycle

קישור לשיחה שלי עם chatGPT:

<https://chatgpt.com/share/6921bd8c-0d08-800a-a63a-4feda942e49b>